

Summary: Dynamic Edge Convolution GNN for IceCube Interaction Localization

GitHub repository: <https://github.com/jlvi179/Advanced-Deep-Learning>

What the Notebook Does

This notebook trains a DynamicEdgeConv-based Graph Neural Network to reconstruct the neutrino interaction position (x, y) from IceCube hit data. It loads the parquet event tables, converts variable-length hit lists into graphs, batches them with PyTorch Geometric, and evaluates the final position predictions with plots and regression metrics.

Architecture Choices

The main design choices are the graph construction, the EdgeConv stack, and the event-level readout:

- **Graph construction:** each hit becomes a node with features such as time, x, y, and optional charge; edges are built dynamically from k-NN neighborhoods in feature space.
- **DynamicEdgeConv layers:** the model recomputes neighbors at every layer, applies an MLP to $[x_i, x_j - x_i]$, and learns local geometric patterns that are hard to capture with a static CNN.
- **Pooling and head:** after the final EdgeConv block, global mean pooling compresses the node embeddings into one event embedding, then a small MLP maps that embedding to the 2D output.
- **Batching:** `collate_fn_gnn` turns awkward arrays into `Data` objects and combines them with `Batch.from_data_list`, which keeps variable hit counts per event manageable.

Fine points to watch:

- **Graph size vs. memory:** dynamic k-NN recomputation is expressive but expensive, so batch size and k should be chosen with GPU memory in mind.
- **Training stability:** BatchNorm, residual connections, dropout, and gradient clipping help deeper EdgeConv stacks stay stable.
- **Checkpoints:** save the model state dict together with the normalization parameters so inference uses the same scaling as training.
- **Event preprocessing:** the normalization and denormalization steps must match exactly, otherwise evaluation plots and metrics will drift.

In short, the notebook turns each IceCube event into a small graph, lets the network learn local hit-to-hit relations through DynamicEdgeConv, and then pools those relations into a compact event-level prediction for the interaction position.